

Once you stop learning, you start dying.

—Albert Einstein

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knowledge bases and chatbots to aid in post-silicon ramp-up. Semiconductor engineers face the challenge of rapidly acquiring diverse skill sets as they work on chips for different applications. To streamline this learning process, companies are using AI-driven chatbots to provide engineers with quick access to training materials, data, and insights, helping them ramp up more efficiently.

In the pre-silicon phase, AI is proving invaluable in the realm of safety simulations. In the automotive industry, safety is paramount, and AI simulations are being used to test and verify chip functionality in critical safety scenarios. This ensures that chips meet the stringent safety standards for applications like autonomous vehicles.

Untapped revenue potential

By integrating AI into various aspects of the semiconductor process, companies can improve operational efficiencies, reduce costs, and create new revenue streams. For instance, AI can optimise capital allocation, enhance product portfolio definition, assist in chip design, and streamline supply chain management.

The McKinsey and Company study suggests that embracing AI in semiconductor development could yield significant returns, with potential savings ranging from 28% to 32% of design costs. These findings underscore the compelling business case for AI adoption in the semiconductor industry.

The semiconductor industry is profoundly transforming, with AI and customisation at the forefront of innovation. From ASICs and domain-specific chipsets to software-defined hardware and AI-driven automation, the industry is embracing change to meet the demands of a dynamic and interconnected world. As AI continues to reshape semiconductor design and development, companies that

leverage its power and adapt to the evolving landscape will be well-positioned for success in this exciting era of semiconductor technology.

Field-programmable gate arrays (FPGAs) offer developers the ability to define custom applications and modules within the chip, providing flexibility and adaptability for iterative development and testing. On the other hand, the SoC approach fabricates chips specifically for a particular application, optimising hardware for efficiency and performance. Advancements like five-nanometre technology in SoCs enable more complex computations.

The integration of AI into the semiconductor industry is further driving innovation, optimising hardware, enabling software-defined upgrades, and improving post-silicon knowledge bases. These advancements promise an exciting future for custom applications and chip design.

It becomes evident that we are on the brink of a transformative era. The demand for AI IoT devices is propelling the industry to innovate at an unprecedented pace, requiring increased computational power, domain-specific chipsets, and software-defined hardware. AI's integration into the entire semiconductor lifecycle promises operational efficiencies and significant cost savings.

The future of semiconductor design and technology is being shaped by AI and customisation, and those who adapt will thrive in this dynamic landscape. As we continue to witness the seamless integration of AI into our daily lives, it is clear that the semiconductor industry holds the key to unlocking the limitless possibilities of a more interconnected and intelligent world. **EFY**

This article is based on a tech talk session at EFY Expo 2023, Delhi, by Balvinder Khurana, Senior Engineering Manager System Validation, NXP. It has been transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Journalist at EFY.